
Collective Semantic Memory: Merge, Trust, and Staleness for Peer Memory in Multi-Agent Navigation

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Abstract

1 **Problem.** Collective memory is often treated as monotone: more shared information should help. In cooperative navigation with scarce semantic targets, this
2 intuition fails. Fast global sharing improves target materialization but causes agents
3 to race for the same targets; slow sharing avoids contention but leaves agents with
4 stale or incomplete memories. **Approach.** We instantiate four memory topologies
5 over the same symbolic-semantic substrate: independent private memories, a
6 shared event bus, a central consensus aggregator, and peer-to-peer broadcast with
7 cadence K . We then introduce a minimal *collective semantic memory* (CSM): peer
8 broadcast plus explicit *merge*, *trust*, and *staleness* rules. Q/R/M/C stage rates are
9 used only as measurement instrumentation, not as the contribution of this paper.
10 **Findings.** On a 35,640-run multi-agent sweep, the fixed-cadence peer family
11 exhibits a bottleneck shift: fast broadcast saturates materialization but collapses
12 completion, while slow broadcast has the opposite profile. Mean time-to-success
13 has an interior optimum at $K=8$. On a 3,240-run scarcity benchmark, the minimal
14 CSM strictly dominates five fixed-cadence peer baselines on success rate: CSM
15 reaches 0.617 with 95% CI [0.610, 0.624], above every peer-K upper CI. **Scope.**
16 Experiments are grid-world substrates. The claim is not that this is a complete
17 theory of collective memory; it is a reproducible architecture split showing that
18 merge/trust/staleness are the right explicit control surfaces after cadence alone is
19 exhausted.
20

21 1 Introduction

22 Many multi-agent systems share information either by writing to a single global memory, by sending
23 learned messages, or by periodically exchanging local state. These mechanisms are usually evaluated
24 end-to-end. If success improves, communication is judged useful; if it does not, the reason is opaque.
25 This paper studies a narrower architectural question: when multiple agents maintain semantic memory
26 about places, what should be shared, at what rate, and with what weight?

27 The answer is not “share everything as soon as possible.” In a scarcity setting, where N agents
28 compete for $T < N$ semantic targets, fresh global memory can make all agents select the same
29 target. The memory is then accurate but unhelpful. Conversely, no communication prevents herding
30 but starves agents of useful discoveries made by peers. The design problem is therefore a trade-off
31 between semantic freshness and occupancy contention.

32 We make three contributions.

- 33 1. **Architecture split.** We implement and compare four memory topologies over the same symbolic
34 substrate: independent, shared-bus, central-aggregator, and peer-to-peer broadcast.

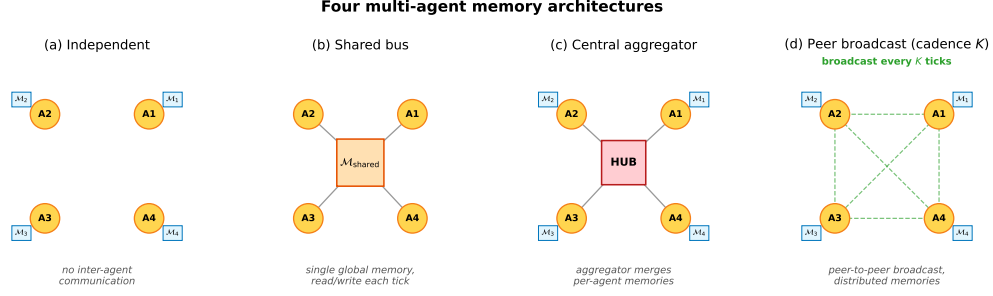


Figure 1: Four memory topologies. **(a)** Independent: per-agent private memory, no exchange. **(b)** Shared bus: one global memory, every agent reads and writes each tick. **(c)** Central aggregator: private memories plus a hub that merges snapshots into a consensus report. **(d)** Peer-to-peer broadcast: private memories with snapshot exchange every K ticks and per-agent merged views.

- 35 **2. Cadence result.** We show that peer broadcast exposes a real coordination parameter K : fast
 36 broadcast improves $P(M^*|R^*)$ but hurts $P(C^*|M^*)$; the efficiency metric has an interior
 37 optimum.
- 38 **3. Minimal CSM.** We instantiate collective semantic memory with three explicit rules: merge,
 39 trust, and staleness. The minimal CSM reaches a new point on the $M \times C$ plane and strictly
 40 dominates fixed-cadence peer baselines on success rate in the scarcity benchmark.
- 41 **Relation to the companion Q/R/M/C paper.** The companion paper treats Q/R/M/C as a diagnostic
 42 framework. Here those stage rates are used as measuring instruments. The main object is different:
 43 the design of distributed symbolic memory itself.

44 2 Memory substrate and task

45 Each agent observes a grid environment and emits semantic place evidence such as `water_source`,
 46 `hazard_edge`, and `open`. A local memory stores place records, tag confidences, visit counts,
 47 freshness, and transition statistics. Planner queries retrieve candidate places by matching required,
 48 preferred, and penalty tags. A selected candidate is materialized into a concrete cell and executed by
 49 a local controller.

50 The multi-agent task uses a 12×10 grid. Each agent must reach a cell tagged `water_source`. Targets
 51 are scarce: $N \in \{3, 5, 8\}$ agents, $T \leq N$ targets, three layouts, three hazard densities, and 40 seeds for
 52 the main cadence sweep. Every run logs Q/R/M/C events per agent:

$$Q^* \rightarrow R^* \rightarrow M^* \rightarrow C^*,$$

53 where M^* means a semantic candidate was locked as an actionable target and C^* means the target
 54 was reached. These logs are not the architectural contribution; they let us see where each topology
 55 fails.

56 3 Four collective-memory topologies

57 The four topologies share the same local memory and planner. They differ only in where merge
 58 happens.

59 **Independent.** Each agent owns a private event store and concept graph. There is no communication
 60 path.

61 **Shared bus.** All observations are appended to a single global event bus. One shared concept graph
 62 is rebuilt from the union of all events.

63 **Central aggregator.** Each agent owns a private concept graph; a central consensus layer pulls
 64 snapshots and computes one trust-weighted report for everyone.

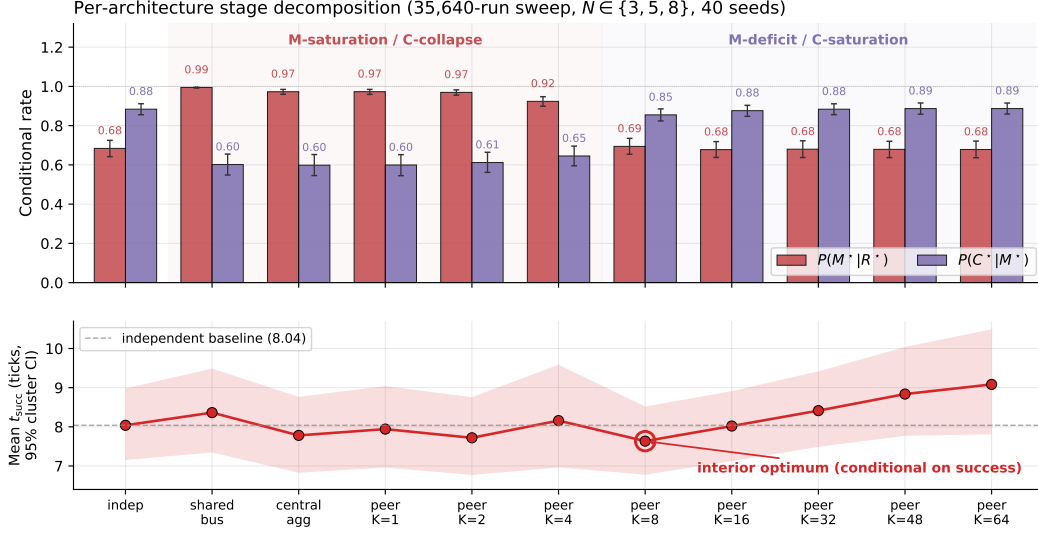


Figure 2: Conditional rates and mean time-to-success from the 35,640-run sweep. Fast sharing improves target materialization but harms completion through contention; slow sharing has the opposite profile. Mean time-to-success has an interior optimum at $K=8$.

65 **Peer broadcast.** Each agent owns a private concept graph and a private merged view. Every K ticks,
 66 agents broadcast snapshots; each agent merges its own memory with the latest snapshot from each
 67 peer.

68 The key distinction is not the clustering primitive. It is the level at which that primitive is instantiated:

independent : N graphs, 0 mergers
 shared bus : 1 graph, 1 implicit global merger
 central : N graphs + 1 central merger
 peer : N graphs + N private mergers.

69 4 Cadence exposes the failure mode

70 The peer topology has one explicit parameter: broadcast cadence K . The main sweep crosses
 71 four architectures and eight peer cadences $K \in \{1, 2, 4, 8, 16, 32, 48, 64\}$ for 35,640 runs. Figure 2
 72 summarizes the result.

73 Fast peer broadcast and centralized sharing live in the M-saturation/C-collapse regime: agents can
 74 lock targets reliably, but many locks point to targets that peers also pursue. Independent and slow
 75 peer broadcast live in the M-deficit/C-saturation regime: fewer agents lock onto good targets, but
 76 those that do are less likely to collide with peer occupancy.

77 Across all peer runs, Spearman of $P(M^*|R^*)$ vs K is -0.608 and of $P(C^*|M^*)$ vs K is $+0.420$
 78 (both $p < 10^{-4}$). The stronger signature is visible at fixed cadence: the within-cadence Pearson
 79 correlation between the M and C links peaks at -0.61 for $K=4$ and decays toward noise as K
 80 grows. Cadence therefore does not simply tune “amount of communication.” It moves the bottleneck
 81 between semantic commitment and shared-resource completion.

82 5 Minimal collective semantic memory

83 The cadence sweep shows that fixed-rate snapshot exchange exposes only one control surface. A
 84 collective semantic memory should expose at least three:

85 **Merge.** How peer evidence about the same semantic place is combined.

86 **Trust.** How evidence from different peers is weighted when it conflicts or becomes unreliable.

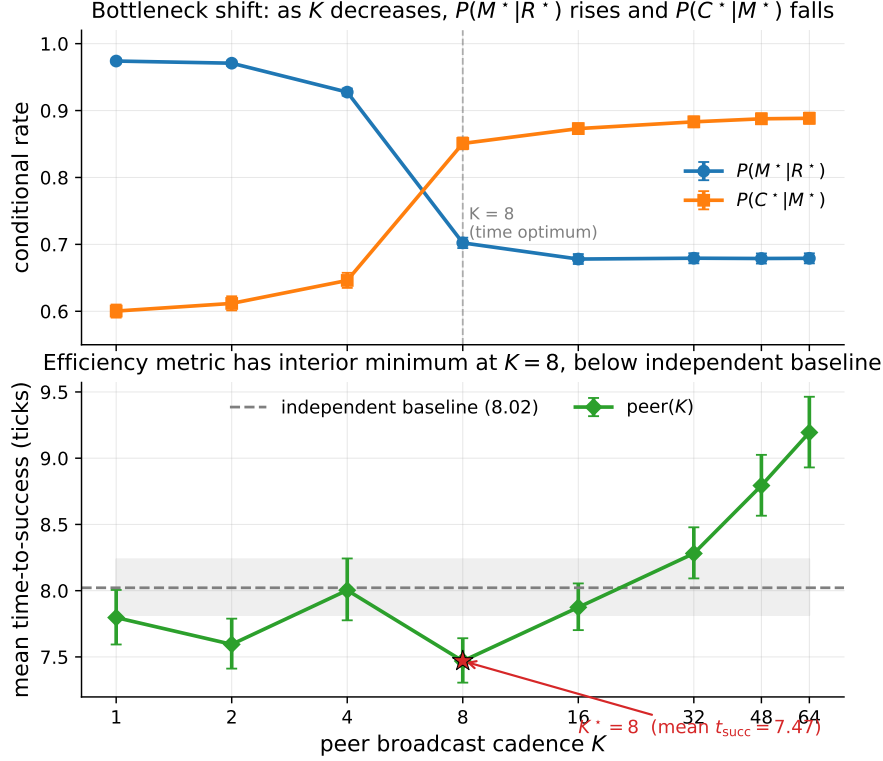


Figure 3: Bottleneck shift across peer cadences. $P(M^*|R^*)$ falls and $P(C^*|M^*)$ rises as K grows. The practical efficiency metric, mean time-to-success, is minimized at $K=8$.

87 **Staleness.** How old evidence decays before it influences retrieval.

88 The minimal CSM uses peer broadcast at $K=8$ and overlays three rules:

89 **Merge.** Per-place score equals own evidence at weight 1 plus peer evidence at weight $\text{trust}_i \cdot e^{-\alpha \cdot \text{age}}$,
90 with $\alpha = 0.05$. Inclusion threshold $\tau = 0.30$.

91 **Trust.** Per-peer EMA on retrieval-success consistency. Peers whose latest snapshot supports the
92 locally locked target gain trust; others decay toward 0.4. Update rate $\beta = 0.10$.

93 **Staleness.** Each broadcast snapshot is timestamped; merge weight decays as $e^{-\alpha \cdot \text{age}}$. Snapshots
94 older than $10K$ ticks are pruned.

95 The implementation is a drop-in replacement for fixed-cadence peer memory:
96 `experiments/collective_semantic_memory/csm_memory.py`.

97 5.1 Results

98 We compare CSM against fixed-cadence peer broadcast at $K \in \{1, 4, 8, 16, 64\}$ on scarcity cells
99 $(N, T) \in \{(3, 2), (5, 3), (8, 5)\}$, three layouts, three hazard densities, and 20 seeds: 3,240 runs total.

100 CSM occupies a third regime. Fast broadcasts have excellent materialization but poor completion;
101 slow broadcasts have the reverse. CSM keeps both conditional links above 0.78. The absolute
102 success gain is modest, +1.7 to +4.0 percentage points, but the stage decomposition shows why it
103 is meaningful: the architecture changes the joint $M \times C$ operating point rather than merely moving
104 along the fixed-cadence curve.

105 **Why CSM wins: the gate, not the sharing.** Provenance annotation of the same runs gives the
106 mechanism. CSM does not share *better* than fixed-cadence peers — it *refuses to act on stale or*
107 *untrusted foreign evidence*. The inclusion rule of Section 5 implies a closed-form horizon beyond

Table 1: Minimal CSM vs fixed-cadence peer broadcast on the scarcity benchmark. CSM’s lower 95% bootstrap CI bound exceeds every peer-K upper CI.

Architecture	$P(M^* R^*)$	$P(C^* M^*)$	Success rate (95% CI)
peer ($K=1$)	0.991	0.586	0.577 [0.572, 0.586]
peer ($K=4$)	0.955	0.623	0.581 [0.572, 0.590]
peer ($K=8$)	0.730	0.871	0.599 [0.591, 0.607]
peer ($K=16$)	0.697	0.892	0.597 [0.589, 0.606]
peer ($K=64$)	0.684	0.900	0.601 [0.594, 0.609]
CSM (minimal)	0.782	0.825	0.617 [0.610, 0.624]

which a peer’s evidence can no longer form an intention, $\text{age}_{\max}(\text{trust}_i) = \alpha^{-1} \ln(\text{trust}_i \cdot c/\tau)$, and this horizon is empirically exact: measured intention-formation cutoffs match the formula tick-for-tick across trust levels (23/23.05, 18/18.59, 12/12.84 at $\text{trust} = 1.0/0.8/0.6$), including a registered-prediction hold-out on six never-run cells (6/6 exact integer breakpoints) and a discrete trust-flip: $\text{trust} \leq \tau/c$ zeroes foreign-evidence commitments entirely. Fixed-cadence peers have no such gate — their foreign evidence never expires — which is precisely why their completion collapses when sharing is fast. The full provenance-conditioned analysis appears in a companion manuscript; here it serves as the mechanistic reading of Table 1.

6 Place identity from meaning: removing the shared frame

Everything above — and the categorical reading below — assumes a shared spatial basis X : ‘the same place’ means ‘the same (x, y) in a frame the environment hands to every agent’. This section removes that assumption and reports what it was hiding. Each agent now holds a *private* frame (the true grid under a secret translation, and, in the final experiment, a secret 90° -multiple rotation); broadcast snapshots carry the sender’s private coordinates, which are meaningless to the receiver until it decides what the sender’s places *mean*.

Mechanism. A *place fingerprint* is the local semantic constellation around a visited cell: which content tags (hazards, walls, water) sit at which relative offsets — translation-invariant and coordinate-free. Two agents match places by *mutually unique* fingerprint correspondence (each side is the other’s sole candidate; one-directional uniqueness aliases distant look-alikes into the receiver’s neighbourhood, and border-only patterns are excluded as descriptions of the world’s shape rather than of a place). The inter-agent frame is the modal translation of ≥ 3 agreeing matched pairs; under rotations, one hypothesis per 90° reuses the same aligner and the winner must dominate the runner-up’s consensus by $2\times$. Foreign evidence — including places the receiver has never seen — is then transported through the recovered frame.

Results. Registered before execution; full protocol in the companion manuscript.

- Exact recovery.** Witness–traveler: the secret offset is recovered exactly in 100% of landmark-world runs, and under full $SE(2)$ frames in 80/80 rotation $\times$ offset $\times$ seed combinations, with 100% completion on foreign-committed locks.
- Failure asymmetry.** Where identification is impossible — a featureless region, or a 180° -symmetric layout in which the frame is unrecoverable *in principle* — semantic identity makes *zero* commitments (fails closed), while coordinate identity keeps confidently locking wrong cells (100% wrong locks; fails open).
- The full stack.** With $N=4$ agents in distinct secret frames, alignment develops online (onset in 100% of episodes, mean tick ≈ 15) and recovers 88% of the shared-frame performance gap (success 0.444 vs. 0.456 oracle vs. 0.350 coordinate); under random $SE(2)$ frames every semantic commitment points at real water (8/8) while the coordinate stack produces 992 phantom locks (0 real).
- The gate is orthogonal.** Re-running the $\text{age}_{\max}$ analysis above with the CSM inclusion rule ON TOP of semantic identity, under a frame offset and under a 90° rotation, reproduces the SAME integer breakpoints (20/14/9; bp = 20 rotated) — the trust $\times$ staleness gate and frame recovery are independent layers.

149 The stronger claim promised in earlier drafts’ limitations is therefore discharged constructively: place
 150 identity can be carried by meaning alone, it composes with merge/trust/staleness unchanged, and —
 151 unlike the coordinate convention it replaces — it knows when it does not know.

152 **The categorical indicator: adjunction round-trip defect.** Frame recovery is exactly the operation
 153 the categorical reading of §7 needs once the shared basis X is dropped: with private frames each agent
 154 carries its own place category, and communication is a pair of recovery functors $F = \text{ALIGN}(A, B)$
 155 (the sender B ’s frame expressed in receiver A ’s coordinates) and $G = \text{ALIGN}(B, A)$. If the two are
 156 genuinely adjoint — the frames are glued — the round trip $\eta_X : X \rightarrow G(F(X))$ is the identity, so
 157 the *adjunction defect*

$$\text{def} = \|F.\text{offset} + G.\text{offset}\|_1 \text{ (translation), } r_F + r_G \equiv 0 \pmod{4} \text{ and } \|R_{r_G} F.\delta + G.\delta\|_1 \text{ (SE(2))},$$

158 is zero (Mac Lane, 1998). Measured on the recovery method above
 159 (experiments/warp/alignment_defect.py, 3 offsets $\times$ 10 seeds per cell): the defect is
 160 *exactly* 0 in 30/30 runs for translation frames and 30/30 for each of the four 90° rotations — the
 161 recovered functors are strict adjoints. Where recovery must fail (an agent that has observed only a
 162 featureless interior band), F or G is undefined in 30/30 runs with *zero* spurious gluings ($\text{def} > 0$):
 163 no adjoint exists and the aligner refuses to build one. The exact discrete aligner therefore realizes a
 164 dichotomy — strict adjoint ($\text{def}=0$) or no adjunction — and never a confident wrong gluing; “failing
 165 closed” above is precisely *refusing a non-adjoint pair*. This is the quantity the categorical model
 166 computes that a re-description does not: a scalar that is zero iff the collective’s local frames are
 167 coherently glued, and it is this gluing — not a shared coordinate convention — that §7’s merge
 168 silently presupposes.

169 7 Categorical BDI reading of collective semantic memory

170 This section gives a compact theoretical reading of the water-search example in terms of belief, desire,
 171 and intention. The goal is not to introduce a new planner, but to make explicit the mathematical
 172 structure already used by the collective semantic memory architecture.

173 We assume a finite set of agents $A = \{1, \dots, N\}$, a shared spatial basis X , and a shared semantic
 174 alphabet Σ . In the present grid-world implementation, X is the common coordinate frame and Σ
 175 contains tags such as `water_source`, `water_candidate`, `hazard_edge`, and `open`. This is an
 176 explicit modelling assumption. The agents may have different observations, different memories, and
 177 different confidence values, but they interpret places in a common coordinate framework and express
 178 semantic evidence in a common tag alphabet. Without the first assumption, peer merge requires a
 179 place-alignment problem to be solved first — Section 6 solves it constructively (fingerprint consensus
 180 recovers the inter-agent frame), and Section 7.1 states what that construction is, categorically. Symbol
 181 alignment (a shared Σ) remains assumed.

182 **What parametrises commonality of interpretation.** It is worth stating explicitly that the cate-
 183 gorical setup localises “how much the agents mean the same thing” in exactly three parameters with
 184 three different statuses. (i) *The alphabet* Σ — what counts as the same *predicate* — is a parameter
 185 of the category itself (the subscript of Ctx_Σ) and is here *assumed*; relaxing it means constructing
 186 a translation between Ctx_{Σ_1} and Ctx_{Σ_2} , which we leave open. (ii) *The alignment morphisms* of
 187 the merge diagram — what counts as the same *place* — were assumed in earlier drafts and are
 188 now *constructed* (Sections 6 and 7.1), with their existence an empirically decidable predicate. (iii)
 189 *The trust $\times$ staleness enrichment* — how much a peer’s interpretation *counts* — makes commonality
 190 graded rather than binary, with its own closed-form horizon and a discrete trust-flip. The three are
 191 ordered by hardness: weights are tunable, place correspondence is solvable from co-observation,
 192 and predicate correspondence is genuinely open. The same triple reappears one level up, at the
 193 evaluator: scoring retrieval and materialization presupposes a correspondence between memory
 194 entities and world referents (see the companion measurement paper’s limitations), and grounding *that*
 195 correspondence by the mechanisms of Section 6 is future work.

196 At time t , agent i maintains a local semantic context $\mathcal{K}_t^i = (G_t^i, \Sigma, I_t^i)$, where G_t^i is the set of place
 197 records stored in the agent’s private memory, and

$$I_t^i \subseteq G_t^i \times \Sigma$$

198 is the thresholded incidence relation saying that place $g \in G_t^i$ carries tag $\sigma \in \Sigma$ with sufficient
 199 confidence. The graded implementation uses tag confidences and weighted scores; the crisp relation
 200 I_t^i is the thresholded skeleton needed for the theoretical description.

201 This formal context induces a Galois connection between sets of tags and sets of places:

$$\text{ext}_t^i(B) = \{g \in G_t^i : \forall \sigma \in B, (g, \sigma) \in I_t^i\}, \quad \text{int}_t^i(S) = \{\sigma \in \Sigma : \forall g \in S, (g, \sigma) \in I_t^i\}.$$

202 For a water-search task, the desired semantic predicate is $B_{\text{water}} = \{\text{water_source}\}$, or, in
 203 the implemented weighted form, a predicate combining positive evidence for `water_source`,
 204 `water_candidate`, and `water_nearby` with penalties for `hazard_edge` or nearby risk. Retrieval
 205 succeeds for agent i at time t exactly when

$$\text{ext}_t^i(B_{\text{water}}) \neq \emptyset.$$

206 Thus the retrieval stage is not a coordinate lookup. It is the test that the queried semantic meaning
 207 already has a non-empty extent in the agent’s realized memory.

208 We can now define the BDI components.

209 **Belief.** The belief state of agent i is the structured semantic memory induced by $\mathcal{K}_t^i$. Equivalently, it
 210 is the concept lattice $L_t^i = \text{Concepts}(\mathcal{K}_t^i)$, whose nodes pair extents of places with intensions of tags.
 211 A water belief is not the proposition “there is water at coordinate x .” It is the region of L_t^i whose
 212 intension contains the water predicate. If this region is empty, the agent does not merely choose
 213 badly; the relevant semantic object is absent from its current memory.

214 **Desire.** Desire is a state-conditioned semantic predicate over the belief lattice. In the water example,
 215 the agent’s internal state contains a thirst variable. The active desire can be written as

$$D_t^i = \begin{cases} B_{\text{water}}, & \text{if } \text{thirst}_t^i \geq \theta_{\text{on}}, \\ B_{\text{goal}}, & \text{otherwise.} \end{cases}$$

216 Thus desire is not an additional metaphysical object. It is the rule that selects which semantic predicate
 217 should be queried from the current belief structure.

218 **Intention.** Intention is the commitment to one materialized candidate. Once desire has selected
 219 a predicate and retrieval has produced a non-empty extent, the planner applies a satisficing choice
 220 function

$$\chi_t^i : \mathcal{P}(G_t^i) \rightarrow G_t^i \cup \{\perp\}, \quad g_t^{i,*} = \chi_t^i(\text{ext}_t^i(D_t^i)),$$

221 where $\perp$ denotes failure to materialize a target. In Q/R/M/C terms, desire determines the query Q^* ,
 222 belief supports retrieval R^* , and intention corresponds to materialization M^* . Completion C^* is
 223 downstream of this semantic structure: it depends on whether the physical controller reaches the
 224 locked target and whether the target remains available under multi-agent occupancy.

225 This gives a compact categorical reading. Let Ctx_Σ be the category whose objects are semantic
 226 contexts (G, Σ, I) over the fixed alphabet Σ , and whose morphisms preserve semantic incidence. A
 227 local memory trajectory for agent i is a time-indexed diagram

$$\mathcal{K}_0^i \rightarrow \mathcal{K}_1^i \rightarrow \dots \rightarrow \mathcal{K}_t^i \rightarrow \mathcal{K}_{t+1}^i \rightarrow \dots$$

228 in Ctx_Σ . Each arrow is induced by observation and memory update. When new evidence is
 229 consolidated, the formal context changes; therefore its concept lattice changes. This is the order-
 230 theoretic content of memory growth.

231 In Q/R/M/C terms, an empty extent $\text{ext}_t^i(D_t^i) = \emptyset$ is a belief/retrieval failure, a non-empty extent
 232 with no stable lock is a materialization failure, and a successful lock that cannot be reached is
 233 a completion failure — the semantic and the physical parts of the agent’s state fail in different,
 234 separately measurable ways.

235 The collective case lifts this by one construction. At broadcast times the receiver i holds a diagram
 236 of local memory objects $\mathcal{D}_t^i = (\mathcal{K}_t^i, \mathcal{K}_t^{j_1}, \dots, \mathcal{K}_t^{j_m})$ together with alignment maps, and queries the
 237 merged consensus object $\tilde{\mathcal{K}}_t^i = \text{Merge}_{\text{trust, stale}}(\mathcal{D}_t^i)$, with peer evidence weighted exactly by the
 238 rules of Section 5 ($\text{trust}_{j \rightarrow i} \cdot e^{-\alpha \text{age}_j}$). The same BDI definitions apply to the merged context:
 239 intention is now selected from $\text{ext}_{\tilde{\mathcal{K}}_t^i}(D_t^i)$ rather than from the private extent alone.

240 This explains why communication cadence produces a bottleneck shift. Faster broadcast increases
 241 the freshness of peer evidence. This tends to enlarge or improve the water extent $\text{ext}_{\tilde{\mathcal{K}}_t}(B_{\text{water}})$, so
 242 agents can materialize targets more reliably; in Q/R/M/C terms, this raises $P(M^* \mid R^*)$. But the
 243 same fresh shared belief also synchronizes intentions. Several agents may lock onto the same scarce
 244 water source. The memory is semantically accurate, but the physical target is now contested. This
 245 lowers $P(C^* \mid M^*)$. The bottleneck is therefore physically grounded. It is not a formal artifact of
 246 the decomposition: it arises because semantic belief is shared through memory, while completion is
 247 constrained by occupancy and scarce resources. Collective semantic memory must therefore regulate
 248 not only how much information is shared, but also how it is merged, whose evidence is trusted, and
 249 when stale evidence should stop influencing intention formation.

250 In this sense the “collective” is not a central mind and not a global map but a stabilised family of
 251 peer-indexed processes; the next subsection makes that statement precise.

252 The dual, process-level reading — agents as coalgebras (codata), the collective as a final coalgebra,
 253 communication as depth-one corecursion, and the bottleneck shift as a competition between belief-
 254 and intention-bisimilarity — is developed in Appendix A. It is a lens, not a mechanism: nothing
 255 downstream depends on it, which is why it lives in the appendix.

256 7.1 Colimit merge and its invariants

257 The merge object above can be described more precisely. Working in a timestamp- and provenance-
 258 enriched version of Ctx_Σ , a snapshot merge from agents $1, \dots, N$ is a diagram of local memory
 259 objects plus alignment maps induced by the place-identity relation. The consensus object $\tilde{\mathcal{K}}_t^i$ is the
 260 colimit of this diagram, implemented concretely by the union-find merge over aligned concepts. This
 261 reading gives three useful constraints.

- 262 1. **Order independence.** The merged object is unique up to isomorphism, so the result is not an
 263 artefact of processing agent snapshots in a particular order.
- 264 2. **Chaining is expected.** If concept A aligns with B and B aligns with C, union-find performs the
 265 transitive closure. Over-merging is therefore not a bug in the data structure; it is the cost of the
 266 chosen equivalence relation.
- 267 3. **Decay is not a morphism.** Staleness is kept as a weight field outside the alignment mor-
 268 phisms. Otherwise age-then-merge and merge-then-age need not commute, breaking the clean
 269 interpretation of broadcast snapshots as one diagram.

270 In earlier drafts this section ended by conceding that the model does not answer which perceptual
 271 features should identify two places — the alignment maps were assumed, not constructed. That gap
 272 is now closed by Section 6, and the closure gives the colimit reading its missing scientific function:

- 273 1. **The alignment morphisms are constructed, not assumed.** Fingerprint consensus IS an
 274 algorithm that produces the alignment maps of the diagram — mutually-unique landmark
 275 matches determine the frame transform through which every local memory object embeds.
- 276 2. **Existence becomes an empirical predicate.** The colimit exists iff the consensus does. In a
 277 180° -symmetric world the alignment morphism does not exist *in principle* (two incomparable
 278 diagrams support equal consensus), and the implementation detects exactly this and refuses to
 279 merge — the fail-closed behaviour of Section 6 is the categorical non-existence made operational.
 280 This predicate is measured, not asserted: the adjunction round-trip defect of §6 is exactly 0 when
 281 the pairwise alignment is a strict adjoint (in every recovered translation and 90° -rotation case,
 282 30/30) and undefined when no adjoint exists (fail-closed, 30/30, with zero spurious gluings) —
 283 a scalar witness for colimit existence.
- 284 3. **Decay stays outside the morphisms** (invariant 3), and empirically the trust $\times$ staleness gate is
 285 unchanged by frame recovery — the same integer breakpoints with and without a shared frame
 286 — confirming that the two layers commute as the diagram reading requires.

7.2 Collective memory as a coherent gluing of local place categories

The colimit merge of §7.1 *assumed* the alignment maps and §6 *constructs* them; we can now state, as the central categorical result, what object the collective memory is when agents hold *different* frames, and exactly when it exists faithfully.

Setup. Agent i 's realized memory is a thin category (a poset): its concept lattice $L_i = \text{Concepts}(\mathcal{K}_i)$ over i 's private frame, objects the places, order the extent inclusion of §7. Frame recovery (§6) yields, on the co-observed subposet, a monotone map $F_{ij} : L_i \rightarrow L_j$ that transports i 's places into j 's frame, and its reverse F_{ji} . Because the categories are *thin*, a recovered pair (F_{ij}, F_{ji}) is an *adjoint pair* precisely when it is a Galois connection, with unit $\eta^{ij} : \text{Id}_{L_i} \Rightarrow F_{ji} \circ F_{ij}$ — the adjunction defect measured in §6.

Definition (coherent alignment system). The family $\{F_{ij}\}$ over the communication graph is *coherent* if (i) each pair is an adjoint order-embedding (η^{ij} an isomorphism, i.e. defect 0: no place is aliased onto a distinct one), and (ii) it satisfies the cocycle condition $F_{jk} \circ F_{ij} = F_{ik}$ on triple overlaps.

Proposition (two agents). For an adjoint pair $F \dashv G$ between L_A, L_B , the collage (comma) category $\text{Gl}(F)$ — objects $L_A \sqcup L_B$, cross-hom $\text{Gl}(F)(X, Y) = L_B(FX, Y) \cong L_A(X, GY)$ (Mac Lane, 1998) — contains L_A, L_B as full subposets and is the two-agent collective memory; the defect η_X is the obstruction to X round-tripping to itself, and the gluing collapses distinct places iff η is not an isomorphism.

Theorem (collective memory = colimit of the alignment diagram). Let $\mathbf{D}$ be the diagram in $\mathbf{Pos}$ (equivalently the enriched Ctx_Σ) whose vertices are the L_i and whose edges are the F_{ij} . The collective memory is $\text{colim } \mathbf{D}$. As $\mathbf{Pos}$ is cocomplete this colimit always exists; it is a *faithful* gluing — every L_i embeds and no two distinct places are identified — *iff* the alignment system is coherent. When all $F_{ij} = \text{id}$ (a shared frame) $\mathbf{D}$ is the constant diagram and $\text{colim } \mathbf{D}$ is exactly the union-find colimit merge of §7.1; the general statement removes the shared-frame assumption by replacing identities with the recovered adjoint embeddings.

Argument. Cocompleteness of $\mathbf{Pos}$ realizes the colimit as the disjoint union quotiented by the preorder generated by the F_{ij} -identifications. Each canonical injection $L_i \hookrightarrow \text{colim } \mathbf{D}$ is faithful iff those identifications add no collapse inside an L_i , which holds iff every F_{ij} is an order-embedding (unit iso, defect 0) and the identifications agree across overlaps (cocycle) — i.e. coherence. The shared-frame case is the constant diagram, whose colimit is the componentwise union, i.e. union-find. $\square$

Corollary (two computable witnesses). Coherence is checkable without solving general semantic alignment. (a) The pairwise *adjunction defect* η (§6) is zero iff each edge is a strict adjoint embedding — measured exactly 0 in 30/30 recovered translation and rotation cases — and guards against *over-merge*. (b) The *connectedness* of the trust $\times$ cadence reachability graph (§7.3, Figure 4) decides whether a single $\text{colim } \mathbf{D}$ spans all N or the diagram disconnects into a coproduct of sub-consensuses (*fragmentation*). The two failure modes are categorically distinct and separately measured: defect > 0 is collapse, disconnection is fragmentation — the two axes of Figure 4 and the fail-closed behaviour of §6.

Scope. The proofs are the standard cocompleteness of $\mathbf{Pos}$ and the collage construction (Mac Lane, 1998); the theorem's job is to name the object and pin its existence to two measured quantities. The empirical content — that fingerprint recovery actually yields the adjoint embeddings — lives in §6, not in the theorem. One assumption is *not* discharged: a shared tag alphabet Σ ; with different alphabets the F_{ij} would have to translate Σ as well, which is beyond the present recovery and is the honest next frontier.

7.3 Protocol parameters as categorical structure: a trust $\times$ cadence phase diagram

The reading so far models the *memory* (objects are semantic contexts, merge is their colimit) under a shared frame and full exchange. Its predictive use comes from turning the lens on the *communication*

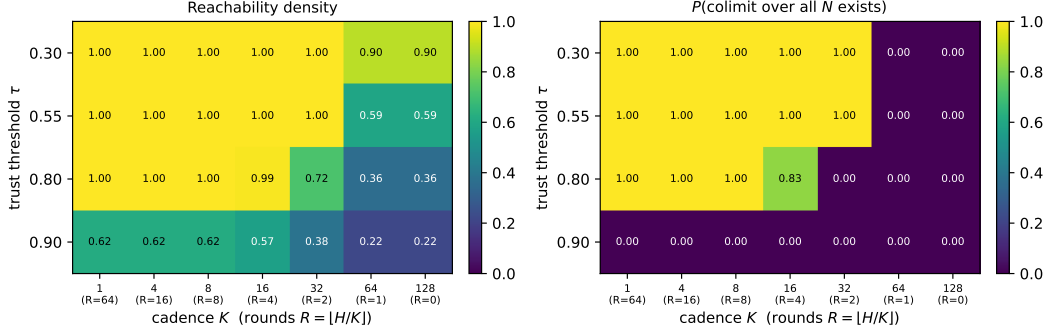


Figure 4: Communication-structure phase diagram over trust threshold τ (rows) and cadence K (columns; $R = \lfloor H/K \rfloor$ broadcast rounds), $N=8$ agents on an interaction ring, 12 seeds. **Left:** reachability density (fraction of ordered agent pairs whose evidence can compose within R rounds). **Right:** P (a colimit over all N exists), i.e. all-to-all reachability. The diagonal boundary is a structural transition: the collective admits a single consensus object in the low- τ /low- K region and fragments in the high- τ /high- K corner, with higher τ demanding lower K (more rounds) to stay connected.

structure and asking *when the colimit exists* as a function of the two protocol parameters — so that the category computes a structural fact rather than re-describing an algorithm. Two parameters play two structural roles:

Trust τ gates morphisms. An edge $i \rightarrow j$ (agent i incorporates j 's evidence) exists only where i 's trust in j is at least τ . Raising τ prunes the trust graph.

Cadence K gates composition depth. Over a horizon H there are $R = \lfloor H/K \rfloor$ broadcast rounds, so a composed chain $i \rightarrow k \rightarrow \dots \rightarrow j$ can be at most R morphisms long. Low K (many rounds) allows deep composition; high K allows only short paths.

A colimit over all N agents — a consensus object that fuses and serves every agent's evidence — exists iff the R -round reachability of the τ -thinned trust graph is all-to-all.

Figure 4 sweeps $\tau \times K$ for $N=8$ agents whose pairwise trust is produced by the system's own EMA rule (`peer_memory.peer_trust.AsymmetricTrust`) under a ring co-observation topology (neighbours co-observe often and earn high trust; distant agents do not), averaged over 12 seeds (`experiments/collective_semantic_memory/phase_diagram_trust_cadence.py`). The result is a genuine phase transition with a diagonal boundary. At low τ the trust graph is dense and the colimit exists at every cadence. Raising τ prunes the long-range morphisms down to a near-neighbour ring; the collective then keeps a colimit *only if* cadence is low enough for multi-hop composition through still-trusted intermediaries to reconnect it — $\tau=0.55$ holds the colimit down to $R=2$, $\tau=0.80$ needs $R \geq 4$ (the ring diameter), and $\tau=0.90$ fragments throughout. Cadence and trust are therefore not opaque knobs: together they determine a categorical structural invariant, and the phase boundary is exactly where the collective transitions from a single consensus object to fragmented sub-consensuses.

This is the sense in which the categorical model is operational rather than descriptive: each protocol parameter maps to a concrete change in the communication diagram, and the diagram's structural invariant (existence of a colimit) predicts a qualitative regime of the collective. *Scope.* The trust matrix is produced by the real EMA rule under a modelled ring co-observation topology, and “colimit exists” is operationalised as all-to-all R -round reachability; establishing quantitatively that crossing this structural boundary predicts the drop in observed Q/R/M/C performance is the natural next experiment.

7.4 Toward symbol alignment: grounding tag alphabets through shared places

The gluing theorem (§7.2) still assumes a shared alphabet Σ . Dropping it, each agent holds a context (G_i, Σ_i, I_i) over its *own* vocabulary, and the right morphism is no longer a functor over one Σ but an *infomorphism* (Barwise and Seligman, 1997): a pair $f : G_A \rightarrow G_B$, $\hat{f} : \Sigma_B \rightarrow \Sigma_A$ with

369 $I_B(fg, y) \Leftrightarrow I_A(g, \hat{f}y)$. Infomorphisms compose and have colimits (Barwise–Seligman channels),
 370 so the collective memory over heterogeneous alphabets is the *channel core* — the colimit of the
 371 infomorphism diagram — which extends §7.2 verbatim (functors over one Σ become infomorphisms
 372 over the Σ_i ; shared Σ with $\hat{f} = \text{id}$ is the old case).

373 **Grounding mechanism.** Symbols align through shared referents: on the places already anchored
 374 by frame recovery (§6), the co-occurrence of an A -tag and a B -tag is evidence they name the same
 375 feature, and $\hat{f}$ is read off the anchored co-occurrence. (When the alphabets differ so much that
 376 fingerprints themselves cannot seed the anchors, one bootstraps from the alphabet-independent
 377 structural constellations and alternates recovery of f and $\hat{f}$.) The *symbol-alignment defect* def_Σ —
 378 the fraction of anchored (place, y) pairs violating the infomorphism biconditional — is the symbol
 379 analogue of the frame adjunction defect of §6.

380 **Preliminary result.** A controlled study (`experiments/warp/symbol_alignment.py`; $|\Sigma_A|=8$,
 381 120 places, 20 seeds) gives agent B a secret relabelling of A ’s alphabet with observation noise, and
 382 recovers $\hat{f}$ from anchored co-occurrence.

Table 2: Symbol alignment by grounding vs. assuming identical alphabets. Target-ID F1 is identifying
 A ’s target feature from B ’s (translated) evidence. “naive” assumes the alphabets coincide by index;
 “recovered” learns $\hat{f}$; “oracle” is the true map.

noise	transl. acc.	def_Σ	F1 naive	F1 recovered	F1 oracle
0.00	1.00	0.000	0.56	1.00	1.00
0.10	1.00	0.099	0.55	0.92	0.92
0.30	0.89	0.304	0.56	0.69	0.73

383 Assuming identical alphabets fails (F1 ≈ 0.55 : the relabelling sends B ’s target tag to the wrong
 384 feature); grounding recovers the map (translation accuracy 1.00, def_Σ tracking the noise floor) and
 385 reaches oracle-level target identification, robustly down to 15% anchored places (accuracy 0.95).
 386 Coarsening two features into one B -tag bounds the achievable accuracy — there is no faithful
 387 bijection, so the channel core coarsens rather than errs. Symbol alignment is thus not a wholly
 388 open problem but the same grounding-plus-defect recipe one alphabet up; a full treatment (genuine
 389 incommensurability, online drift) is the next paper.

390 8 Limitations and next experiments

391 **Grid substrate.** The current result is intentionally controlled: all experiments are in grid worlds. A
 392 continuous-state substrate such as VMAS, Habitat, ProcTHOR, or MiniWorld is the next portability
 393 step.

394 **Coordinate anchor.** Removed in Section 6: local fingerprint signatures recover the inter-agent frame
 395 (translation and 90° rotations) exactly, and the merge/trust/staleness results are unchanged on top
 396 of the recovered frames. What remains open is the harder end of the same axis: continuous $SE(2)$,
 397 appearance noise (the current matcher uses exact constellation equality because the substrate is
 398 noise-free). Symbol alignment — agents with *different tag alphabets* — is addressed preliminarily in
 399 §7.4 (grounding through shared places, recovered as an infomorphism); genuine incommensurability
 400 and vocabulary drift remain open.

401 **Minimal CSM.** The CSM rules are deliberately simple. Adaptive cadence K_t , intent-conditioned
 402 trust, and selective consolidation of under-represented tags are the next architecture variants.

403 **Learned baselines.** CommNet-style learned communication is a useful comparison, but the more
 404 relevant future baseline is an explicit memory LLM or vector-memory multi-agent system measured
 405 on the same task.

9 Conclusion

Collective memory is not monotone. In scarce multi-agent navigation, instant information sharing can reduce completion by causing agents to converge on the same targets. Peer cadence exposes the trade-off, but cadence alone is a one-dimensional control surface. A collective semantic memory should make merge, trust, and staleness explicit. The minimal CSM does exactly that and moves the system to a new operating point: neither fast-broadcast M-saturation nor slow-broadcast C-saturation, but a more balanced $M \times C$ profile with higher success than every fixed peer cadence in the scarcity benchmark.

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A The coalgebraic layer: individuals as codata, the collective as a final coalgebra

The reading of §7 fixes belief *content*: at a state, the concept lattice $L_t^i = \text{Concepts}(\mathcal{K}_t^i)$ says what is retrievable, and the merge of §7.1 says how peer contents combine into $\tilde{\mathcal{K}}_t^i$. Both are *algebraic*: they build a belief object. Neither speaks to the agent as a non-terminating *process* that must emit an action and a signal at every tick under a moving environment, nor to the sense in which a *population* comes to share a convention. That process side is *coalgebraic* (Rutten, 2000; Jacobs, 2016). We make it precise on the water-search example and show the two sides are dual, meeting inside a single state transition. (The belief/desire/intention vocabulary follows the BDI agent tradition (Rao and Georgeff, 1995), here given coalgebraic content.)

452 **Signature.** Fix an action set Act (move, dig, noop), a per-agent observation set O_i (agent i 's local
453 percept — its *unique view*), and an *ostension* set

$$M = 1 + (X \times \Sigma),$$

454 whose values are either *silence* (the summand 1) or a pointer “place x carries tag σ ” (whistle-and-point
455 at a water-marked cell). That silence is a first-class value of M matters: the ostension channel *can be*
456 *informatively empty*, and it is separate from the state update below. This is exactly the observability
457 that a learned real-valued message vector destroys (its channel is never empty and never separates
458 signalling from stepping), and it is why the symbolic stages remain readable.

459 **State.** We take the coalgebra carrier to be the H-MDP high-level state of §7,

$$\mathcal{S}_i = \{ h = (\mathcal{K}^i, D^i, g^{i,*}, s^i) \},$$

460 carrying the same information as $(L^i, D^i, g^{i,*}, s^i)$ since $L^i = \text{Concepts}(\mathcal{K}^i)$. Thus *belief lives in*
461 *the state*; the lattice is a projection of it.

462 **Definition (codata: the agent's cotype).** The observable behaviour of agent i is a coalgebra for the
463 endofunctor

$$B_i(S) = \underbrace{\text{Act}}_{\text{act now}} \times \underbrace{M}_{\text{signal now}} \times \underbrace{S^{O_i \times M^{A \setminus \{i\}}}}_{\text{continue given (my next percept, peers' signals)}},$$

464 i.e. a map $c_i = \langle \text{observe_act}_i, \text{observe_ostension}_i, \text{step_intension}_i \rangle : \mathcal{S}_i \rightarrow B_i(\mathcal{S}_i)$
465 with three destructors

$$\text{observe_act}_i : \mathcal{S}_i \rightarrow \text{Act}, \quad \text{observe_ostension}_i : \mathcal{S}_i \rightarrow M, \quad \text{step_intension}_i : \mathcal{S}_i \rightarrow S^{O_i \times M^{A \setminus \{i\}}}.$$

466 Here $M^{A \setminus \{i\}} = \prod_{j \neq i} M$ is the tuple of peers' ostensions delivered this tick. Operationally,
467 $\text{step_intension}_i(h)(o, (m_j)_{j \neq i})$ (i) folds the percept o and the peer pointers (m_j) into the context
468 by the trust/staleness merge, $\tilde{\mathcal{K}}_t^i = \text{Merge}_{\text{trust, stale}}(\dots)$ of §7.1; (ii) recomputes the lattice; (iii) re-
469 evaluates the desire predicate D^i ; and (iv) re-locks the intention $g^{i,*} = \chi_t^i(\text{ext}(D^i))$ of §7. So Merge
470 and χ are *components of one destructor*. The agent is the codata generated by c_i : it has no finite
471 terminal plan, only the standing ability to emit an action and an ostension and to continue. This is the
472 “infinite survival generator” of the motivating example.

473 **Definition (indexed family).** The system is the A -indexed family $\{(\mathcal{S}_i, c_i)\}_{i \in A}$, a coalgebra in
474 Set^A . The functor B_i depends on the index i through O_i (and through the peer set $A \setminus \{i\}$): the
475 same survival cotype is instantiated at each agent through that agent's local viewpoint. Agent 1 on
476 the dune generates from “I see the horizon”; agent 2 in the hollow generates from “I feel the soil.”
477 The index is the viewpoint.

478 **Definition (codependence: the collective coalgebra).** Let the environment deliver a joint percept
479 $o = (o_i)_{i \in A} \in \prod_i O_i$. The collective one-tick map on $\prod_i \mathcal{S}_i$ wires each agent's ostension output
480 into the others' step inputs,

$$\Phi((h_i)_i)(o) = \left(\text{step_intension}_i(h_i)(o_i, (\text{observe_ostension}_j(h_j))_{j \neq i}) \right)_{i \in A}.$$

481 Because i 's next state is a function of the *destructor outputs* of the others, the family is not N
482 independent processes but one coalgebra for the product functor $\mathbf{B}(S) = \left(\prod_i (\text{Act} \times M) \right) \times S^{\prod_i O_i}$
483 on $\prod_i \mathcal{S}_i$. This wiring is the *codependence*: agent 1 cannot step without passing agent 2's ostension
484 through its own step_intension . Broadcast cadence enters here as a gate: at non-broadcast ticks
485 the delivered tuple (m_j) is the last received (stale) tuple, or silence; K is the delivery period.

486 **Definition (behaviour and bisimulation).** B_i is a container (polynomial) functor, so it has a final
487 coalgebra $(\nu B_i, \text{beh}_i)$ (Aczel, 1988; Rutten, 2000) whose elements are the $(O_i \times M^{A \setminus \{i\}})$ -branching,
488 $(\text{Act} \times M)$ -labelled behaviour trees, and there is a unique coalgebra morphism $!_i : \mathcal{S}_i \rightarrow \nu B_i$ sending
489 a state to its whole future behaviour. A B_i -*bisimulation* is a relation $R \subseteq \mathcal{S}_i \times \mathcal{S}_i$ under which
490 related states emit equal action and equal ostension and, on every input, step to R -related states.
491 Container functors preserve weak pullbacks, so the *greatest bisimulation* $\sim_i$ exists, is an equivalence,
492 and equals $\ker(!_i)$: two states are bisimilar iff they have the same behaviour (Sangiorgi, 2011). The
493 same holds for the collective coalgebra $(\prod_i \mathcal{S}_i, \langle (\text{observe_act}_i)_i, (\text{observe_ostension}_i)_i, \Phi \rangle)$,
494 with greatest bisimulation $\approx$.

Definition (society / culture). The *collective* is the image of the states in the final coalgebra $\nu\mathbf{B}$, equivalently the quotient $(\prod_i \mathcal{S}_i) / \approx$. It is neither a central server nor a shared map: it is a bisimilarity class of joint behaviour. A *water-search convention* is this invariant read on the water sub-signature. Let $M_{\text{water}} \subseteq M$ be the pointers whose tag lies in the water family, and let $\nu B_i \rightarrow \nu B^w$ be the forgetful map to the behaviour trees restricted to receiving M_{water} and emitting the corresponding intention lock. The population *holds the convention* when all agents’ reachable behaviours agree under this map: hearing “point-and-dig at a water-marked place” drives every agent’s intention destructor to lock that place (up to occupancy). Governance is then automatic in both directions. A new agent (the example’s fourth agent) is a member of the society exactly when its behaviour lands in the class $[\cdot]_{\approx}$ on M_{water} — otherwise its ostensions are not understood and its own step never converges (top-down); and the class itself was reached by the local, coupled steps of the first agents at one green shrub (bottom-up).

Remark (communication = lazy, depth-one corecursion). The behaviour $!_i(h)$ is defined by unfolding c_i indefinitely. On receiving $m_j = \text{observe_ostension}_j(h_j)$, agent i uses only this *depth-one* destructor output inside step_intension_i ; it does *not* compute $!_j(h_j) \in \nu B_j$, the peer’s full behaviour, which is the “I think that you think...” regress. Communication is therefore corecursion truncated to the least depth that resolves the receiver’s own next step — a finite approximation of the final-coalgebra map — with a relevance-theoretic stopping rule (Sperber and Wilson, 1995) deciding that depth. The regress collapses into one effective action.

Remark (algebra–coalgebra bridge to §7–7.1). Belief is algebraic: the FCA lattice L_t^i built by the colimit merge is the *content at a state*. The agent is coalgebraic: the codata observed by the three destructors is the *process over states*. They are dual and meet in one place — the colimit merge and the satisficing choice χ are the internals of step_intension . The memory-growth diagram $\mathcal{K}_0^i \rightarrow \mathcal{K}_1^i \rightarrow \dots$ of §7 is precisely the belief-projection of the coalgebra trajectory $h_0^i \mapsto h_1^i \mapsto \dots$.

Coalgebraic reading of the bottleneck shift. The result of §4 and Table 1 has a one-line reading here. Two kinds of bisimilarity compete. Bisimilarity on the *ostension / belief* sub-signature is the wanted invariant: it is the shared water convention. Bisimilarity on the raw *intention* output over a *scarce* target is the unwanted one: several states mapping to the same g^* is a collision. Occupancy $\mathcal{O}_{-i}$ is the extra-coalgebraic constraint that penalises the second. Cadence K tunes how hard the coupling Φ drives the collective toward the *full* greatest bisimulation $\approx$: small K delivers fresh ostensions often, enlarging the merged water extent so intention locks reliably ($P(M^* \mid R^*) \uparrow$), but over-synchronises the intention destructor so agents collide ($P(C^* \mid M^*) \downarrow$). The interior optimum $K=8$ is where belief converges without intention collapsing onto one cell; the minimal CSM adds trust and staleness precisely so that step_intension reaches belief-bisimilarity (fresh, trusted, aligned evidence) while damping intention-bisimilarity (stale evidence decays before it can herd every agent onto the same lock). We state this as a reading consistent with the data, not as a theorem: the container functors, their final coalgebras, and bisimulation are standard constructions used here as a lens, and the relevance-theoretic stopping rule is an interpretive gloss rather than a fitted mechanism.

533 B Context is not memory: raw context vs. the Songlines graph under one 534 model

Modern LLMs *accept* long context; the question this appendix measures is whether they turn it into a stable, globally coherent state — and whether the Songlines graph supplies that state instead. In every arm the *same* model answers; only the memory substrate differs: **raw** (the full observation transcript in the prompt — typical usage) vs. **graph** (the same evidence consolidated by the Songlines rule: latest state per place, staleness explicit, $O(\text{salient places})$ prompt). Three axes (experiments/context_vs_structure/; HF backend on SLURM GPU nodes and local Ollama as an independent second backend).

Axis 1: volume and integration (one witness log). A witness log over a 30×24 grid contains a freshness conflict (water seen repeatedly at site A , later dried; one fresher sighting at B). Questions: *fact* (where is water now), *count* (how many waters now), *hop* (nearest hazard to current water).

Two failure layers, both with identified mechanisms. *Cognitive*: the raw arm loses global state at **any** volume — on *count* it answers “2” (17/20) or “3” (3/20) when the truth is 1: it adds up historical

Table 3: The motivating example’s terms as formal objects of the coalgebraic reading. Each row is the water-search meaning of a construction already used in §A.

Informal term	Formal object	Water-search meaning
Cotype (codata)	B_i -coalgebra (S_i, c_i)	the agent’s non-terminating action/signal generator
Indexed type	family over A ; dependence of B_i on O_i	the agent’s unique local viewpoint (dune vs. hollow)
Codependent type	wiring $\text{observe_ostension}_j \rightarrow \text{step_intension}_i$ in Φ	“I step only through your ostension”
Individual (instance)	a state $h \in S_i$ and its behaviour $!_i(h)$	the running agent digging and whistling now
Destructors (cogenerators)	observe_act , observe_ostension , step_intension	step-and-dig; whistle-and-point; fold neighbour’s signal into next state
Lazy evaluation	depth-one corecursion + relevance stop	unfold the peer’s coalgebra one step, not its whole behaviour
Final coinductive core / society	νB ; greatest bisimulation $\approx$ (on M_{water})	the stable, distributed water-search culture

Table 4: Accuracy by question, arm, and log volume L (lines). Qwen2.5-7B (cluster, 20 seeds/cell) on cells within its compute envelope; Llama-3.1-8B (local, 10 seeds/cell) at large L with *verified full ingestion* (prompt_eval_count=25,030 tokens at $L=2000$; no truncation). Graph prompts stay ≈ 350 bytes at every L .

Question	Arm	$L=50$	$L=200$	$L=800$	$L=2000$
fact	raw	1.00/–	1.00/–	–/0.00	–/0.00
fact	graph	1.00/–	1.00/–	1.00/1.00	1.00/1.00
count	raw	0.00 /–	0.00 /–	–/ 0.00	–/ 0.00
count	graph	1.00/–	1.00/–	1.00/1.00	1.00/1.00
hop	raw	0.00/–	0.05/–	–/0.00	–/0.00
hop	graph	0.00/–	0.15/–	0.40/0.50	0.30/0.60

547 mentions instead of resolving freshness, i.e. no consolidation. At large L even single-fact retrieval
548 collapses despite verified full ingestion (a probe with the answer at the very end of a 25k-token
549 prompt is also missed). *Systemic*: on the cluster’s 24 GB sm75 GPUs, raw-arm inference at 5–10k
550 tokens exhausts memory through the quadratic attention matrix (marked as compute-envelope cells,
551 “–”, never reported as cognition), while the graph arm runs at constant cost at every volume. Raw
552 context loses twice — as cognition and as systems cost; *hop* shows the honest boundary: structure
553 repairs retrieval, not arithmetic reasoning.

554 **Axis 2: time (sessions).** A persistent world drifts (water relocates, $p=0.35$ /session); after each of 8
555 sessions the model is asked where the water is now, from **raw replay** (all past transcripts re-injected,
556 the typical chat pattern) vs. the persisted **graph**. Qwen2.5-7B: the graph arm holds 0.62–1.00 across
557 all eight sessions at a flat ≈ 0.3 kB prompt; the replay arm’s valid cells are already at 0.00–0.67 in
558 the earliest sessions and its prompt grows to 25 kB by session 8 (later cells envelope-limited on the
559 cluster; on the local backend, where no envelope exists, replay degrades to 0.00–0.50 by 19 kB —
560 the same pattern).

561 **Axis 3: system size (multi-agent, private frames).** Three agents accumulate observations over G
562 sessions, each logging in its *own* secret coordinate frame. Question: “how many DISTINCT water
563 sources exist?” — the same physical water seen by two agents at different private coordinates must
564 be counted once. The raw arm gets all three logs (and is told the frames differ); the graph arm gets
565 the Songlines machinery’s merged map (fingerprint frame recovery, §6, plus dedup). Qwen2.5-7B:
566 graph accuracy *rises* with growth — 0.40, 0.90, 1.00, 1.00 at $G=1, 2, 4, 8$ (≈ 330 bytes) — because
567 frame alignment locks as fingerprints accumulate (the exact% curve of the growth study); the raw arm
568 (13k–105k chars) does not produce a single correct answer on any envelope-valid cell. Cross-frame
569 identity is precisely what a context window cannot represent and the gluing structure can.

570 **Summary.** The same model, given the same evidence, preserves long, drifting, multi-agent context
 571 when — and only when — the evidence reaches it as the consolidated Songlines structure: flat-cost
 572 prompts, freshness resolved, frames glued. Given the raw stream instead, it fails on integration at
 573 1.3 kB, on freshness at 19 kB with full ingestion, and on cross-frame identity at every tested size.
 574 Scope: two model families, one task family per axis, and the compute-envelope cells are reported
 575 as such; the graph arm’s consolidation here is the deterministic Songlines rule — LLM-driven
 576 consolidation is the roadmap’s next stage.

577 C Reproducibility commands

- 578 • Main cadence sweep: `pythonexperiments/big_experiment/exp_cadence_phase.py--modefull--workers8--out_dirtmp/big_experiment_qrmc_40`
- 579
- 580 • Extended cadences: `pythonexperiments/big_experiment/exp_extra_K.py--workers8--out_dirtmp/big_experiment_extraK`
- 581
- 582 • Q/R/M/C analysis: `pythonexperiments/big_experiment/analyze_qrmc.py--runs_csvtmp/big_experiment_qrmc_40/runs.csv--out_dirtmp/big_experiment_qrmc_40`
- 583
- 584
- 585 • Minimal CSM benchmark: `python-mexperiments.collective_semantic_memory.run_csm_benchmark--out_dirtmp/csm_benchmark`
- 586
- 587 • MiniGrid portability sweep: `python-mexperiments.minigrid_multiagent_wrapper.run_portability_sweep--out_dirtmp/minigrid_portability`
- 588
- 589 • Adjunction round-trip defect of frame recovery (Section 6): `pythonexperiments/warp/alignment_defect.py`
- 590
- 591 • Context-vs-structure suite (Appendix B): `experiments/context_vs_structure/run_ctx_vs_struct.py`, `run_persistence_llm.py`, `run_growth_llm.py` (HF backend on GPU or --backend ollama locally)
- 592
- 593
- 594 • Trust×cadence structural phase diagram (Section 7.3, Figure 4): `pythonexperiments/collective_semantic_memory/phase_diagram_trust_cadence.py--figfigures/fig_phase_diagram.pdf`
- 595
- 596

597 D Asset and license note

598 The experiments use MiniGrid and MiniWorld (Chevalier-Boisvert et al., 2023). Locally implemented
 599 benchmark and memory code will be released under the MIT License as part of the artifact package.

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Justification: The paper describes a unified benchmark runner and source-of-truth artifacts. An anonymized public release of the codebase — single-agent benchmark, multi-agent grid environment, 35,640-run sweep driver, Q/R/M/C analyser, oracle-intervention scripts, and exact commands to reproduce every table and figure — accompanies the submission as supplemental material (anonymized GitHub repository; single-command reproduction of the main multi-agent sweep on an 8-core machine in ≈ 22 minutes, matching Appendix H).

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